MIL-D-45470 (Ord)
28 March 1960

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, T76
METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of detonator designated as Detonator, Electric, T76.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111 - Tape, Paper, Gummed (Kraft)
LLL-B-631 - Boxes, Fiber, Corrugated (For Domestic Shipment)

MILITARY

JAN-P-120 - Packaging and Packing for Overseas Shipment-Cartons, Folding, Paperboard
MIL-G-2550 - General Specification for Ammunition Except Small Army Ammunition.

FSC 1375

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

ORDNANCE CORPS

8826685 - Electric Detonator, T76, Assembly

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes
IEL-8792231 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on Drawing (dwg.) 8826685, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be free of chips, dirt, rust, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The quality assurance provisions of this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. Inspection shall be performed in accordance with this specification, Specification MIL-G-2550 and other specifications referenced in any of the contractual documents.

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4.1.1 Contractor inspection.-Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to ber performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.3 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with allquality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor.

f. Number of items in the lot.

g. Data submitted.

The certificate shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of the agent's authority to bind his principal.

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4.1.4 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied, and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with.

4.1.4.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.4.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.1.4.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of metal parts produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting performance, safety or fit, as determined by the Government, shall not be cause for changing the lot number.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.3.

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4.2.2.1 Cup (see dwg. 8826682 covering a detail of dwg. 8826685).

Categories and defects	Method of inspection
Critical	
a. Split, cut or crack through cup	Visual
Major AQL 0.65 percent	
1. Inside diameter, minimum (min.)	Gage
2. Outside diameter, maximum (max.)	Gage
3. Total length, min	Gage
4. Deep scratch or groove in cup	Visual
5. Foreign matter	Visual
6. Cup dented or distorted	Visual
7. Thickness through bottom	Gage
Minor AQL 1.00 percent	
41. Radius incorrect	Visual
42. Evidence of poor workmanship (see 3.3)	Visual

4.2.2.2 Plug (see dwg. 8826684 covering a detail of dwg. 8826683).

Categories and defects	Method of inspection
Major AQL 0.40 percent	
1. Large outside diameter, min	Gage
2. Length of plug	Gage
3. Small outside diameter	Gage
Minor AQL 1.5 percent	
41. Length to shoulder	Gage
42. Chamfer missing or inadequate	Visual
43. Foreign matter	Visual
44. Evidence of poor workmanship	Visual

4.2.2.3 Plug assembly (see dwg. 8826683 covering a detail of dwg. 8826687).

Categories and defects	Method of inspection
Major AQL 0.40 percent	
1. Resistance between plug and pin	Test (see 4.3.1)
2. Push test failure	Test (see 4.3.2)
3. Pin not separated from plug 360° by bonding material	Visual
Minor AQL 0.40 percent	
41. Evidence of poor workmanship (see 3.3)	Visual

4.2.3 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment No. 8792231, or approved equivalents thereto. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, andutilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages,

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standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Resistance between plug and pin.-A direct current (d.c.) of 325 minus 50 volts shall be applied to the plug assembly using the test equipment specified in 4.2.3. The resistance shall be measured to determine compliance with the drawing requirement.

4.3.2 Push test of pin.-The axial force specified on the drawing shall be applied to the contact end of the pin. Observation shall be made to determine that the pin withstands the force applied without any indication of the pin slipping within the plug.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The inner cartons shall be constructed of 130 pound test paperboard complying with Style III, Type B, Class a, Specification JAN-P-120. The inside dimensions of the cartons shall be 7 1/2 by 7 1/2 by 3 1/4 inches. Any void space shall be filled with plain kraft paper to insure a tight pack.

5.1.1.1 Cups.-Ten thousand (10,000) cups plus average shall be packaged in a carton constructed as specified in 5.1.1.

5.1.1.2 Plug assembly.-Ten thousand (10,000) plug assemblies plus average shall be packaged in a carton constructed as specified in 5.1.1.

5.2 Packing.

5.2.1 Level C.-The outer carton shall be constructed of 200 pound test corrugated fiberboard complying with Style RSC, Specification MIL-B-631 except that the body joints of the carton shall be secured with metal staples or metal stitches regularly spaced not more than an average of 1 1/2 inches, center to center.

5.2.1.1 Cup and plug assembly.-Ten thousand (10,000) cups and ten thousand (10,000) plug assemblies plus average in paperboard cartons (see 5.1.1.1 and 5.1.1.2) shall be packed in a corrugated fiberboard carton constructed as specified in 5.3. The inside dimensions of the carton shall be 7 3/4 by 7 3/4 by 7 1/2 inches.

5.3 Marking.-Marking of the containers shall be as specified in Standard MIL-STD-129.

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5.3.1 Exterior containers inclosing packages prepared for Level C shall be conspicuously marked MIN-MIL-PK.

5.4 Packing requirements.-All carton flaps shall meet when closed. A manufacturing tolerance of plus or minus 1/16 inch shall apply to all packing material. The parts vendor shall be furnished preproduction samples of all packing materials so that proper fit of packing materials and metal parts can be determined. Minor modifications of packing material dimensions to provide a proper fit of all parts, as determined to be necessary by examination of sample packs, are permissible. Additional filler material may be used as necessary to insure a tight pack.

5.5 Sealing.-No sealing required for the inner cartons. The outer carton flaps shall be secured with gummed kraft paper not less than 3 inches wide complying with Federal Specification UU-T-111 for Class 2 and 3 tape. The tape shall be centered the full length of the outer flap seams, and extend over each end of the carton at least 3 inches. The flap edge seams shall be similarly sealed with the tape extending at least 3 inches along the corner of the carton. Tape shall be properly moistened and sufficiently compressed to insure tight adhesion to the carton. Alternatively when acceptable to the installation using the metal parts, flaps may be glued using a suitable adhesive applied over at least 75 percent of the contact area between inner and outer flaps. When flaps are pulled apart after gluing, they shall show complete fiber failure in the glued area.

5.6 Shipping.-When component parts from more than one lot are shipped in the same carrier, each lot must be kept separate and short packed cartons placed in a conspicuous location. The division between the lots must be clearly indicated to prevent mixing of lots in transit.

5.6.1 When feasible, for small and light weight parts, the entire packing lot quantity including the percentage of extra parts shall be packed in the prime container.

5.7 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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